



VISUAL ANALYTICS: INTERACTION SCIENCE

AB MOSCA (THEY/THEM)



TODAY'S PLAN

- Science of interaction
- Define interaction for visualization
- Leveraging interaction

RECALL: WHAT IS VISUAL ANALYTICS?



discovery and **communication** of
meaningful patterns in data

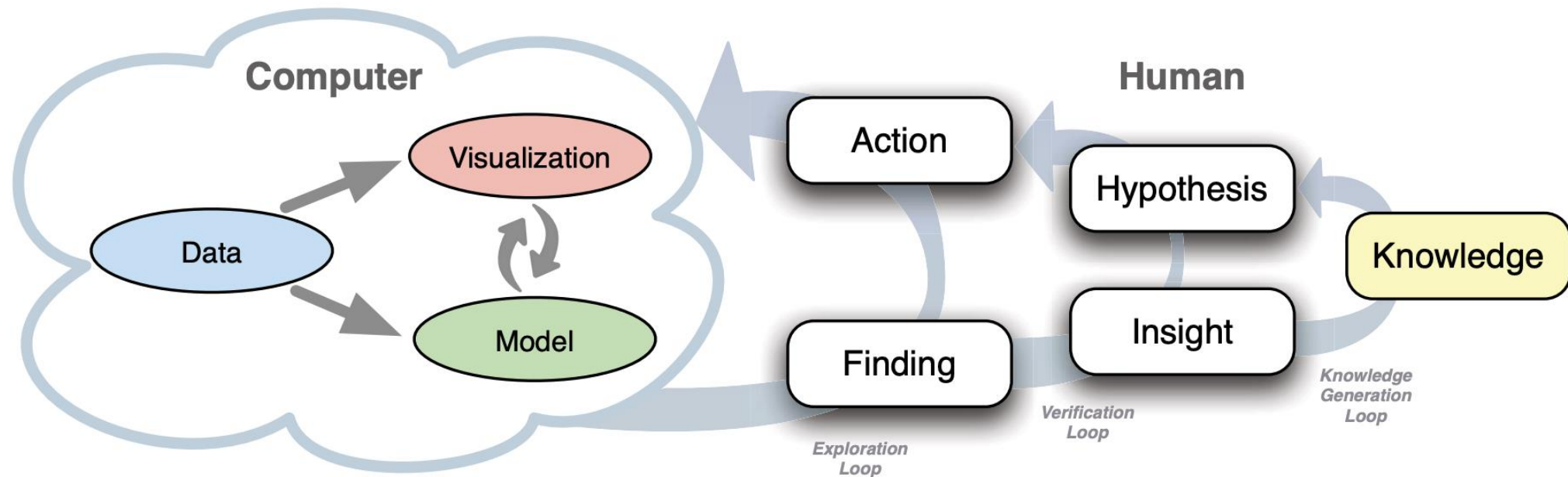
“The science of **analytical reasoning**
facilitated by **interactive visual**
interfaces”¹

¹Thomas, James J., and Kristin A. Cook, eds. *Illuminating the Path:
The research and development agenda for visual analytics*.
Computer Society Press, 2005.

visual **representations** of data
that reinforce human **cognition**

RECALL: WHAT IS VISUAL ANALYTICS?

Big Idea: Leverage complimentary strengths of humans and computers



D. Sacha, A. Stoffel, F. Stoffel, B. C. Kwon, G. Ellis and D.A. Keim, "Knowledge Generation Model for Visual Analytics," in IEEE Transactions on Visualization and Computer Graphics, vol. 20, no. 12, pp. 1604-1613, 31 Dec. 2014, doi: 10.1109/TVCG.2014.2346481.

SCIENCE OF INTERACTION

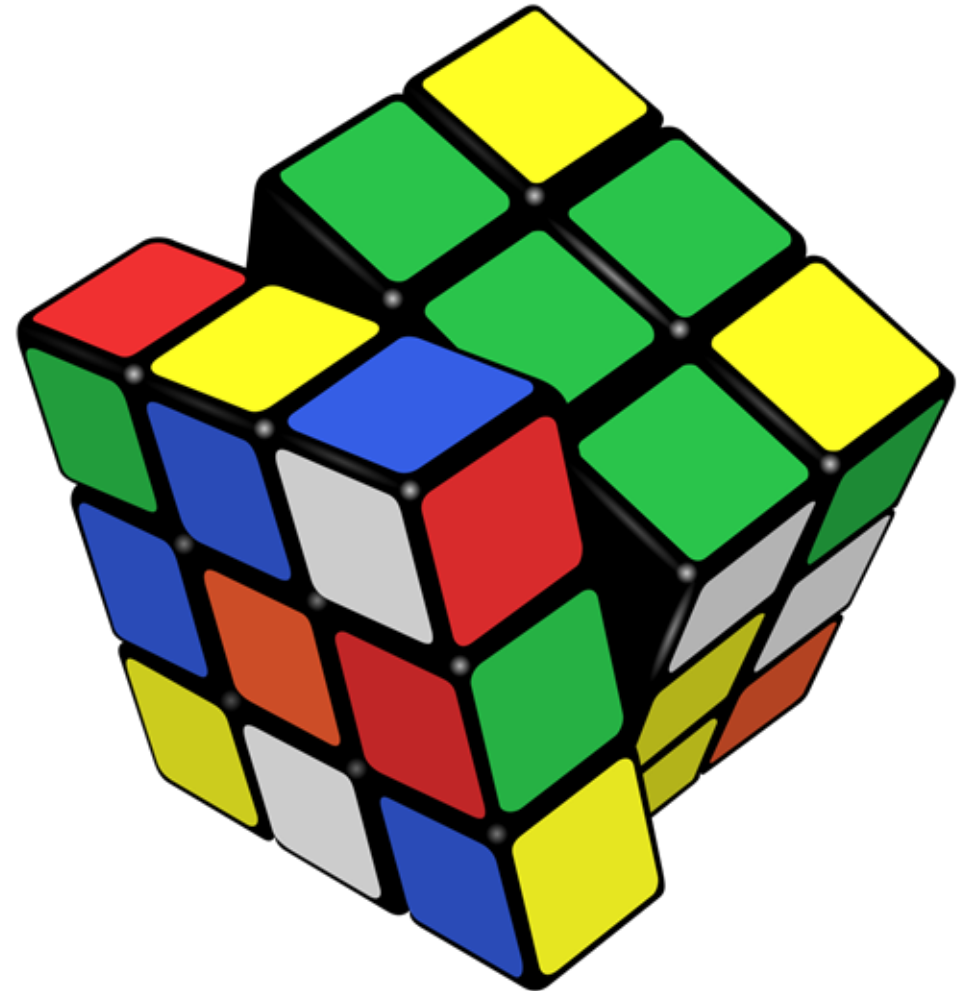
- Science of interaction is the study of methods by which humans create knowledge through manipulation of an interface
- Interaction refers to:
 - Set of controls through which a user can manipulate an interface
 - Cognitive act between the user and the problem space facilitated by computational tools

More concrete
concept

Abstract
concept

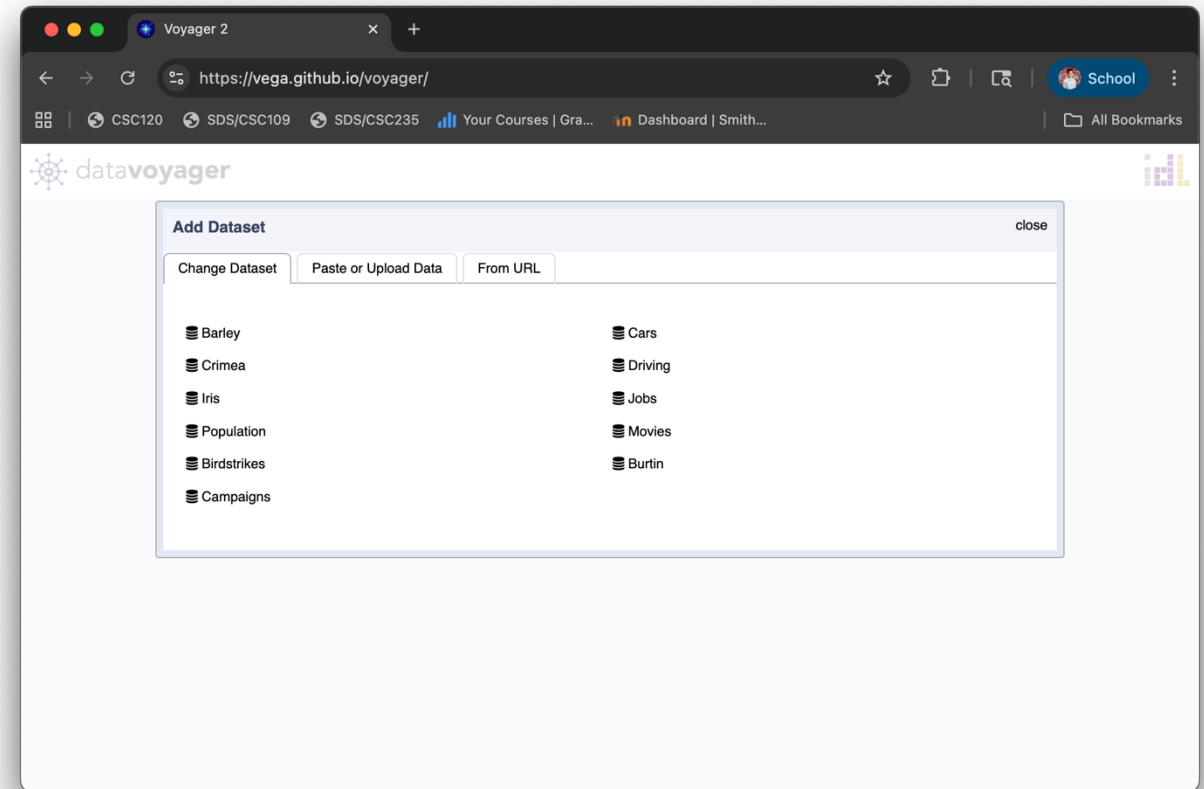
SCIENCE OF INTERACTION

- Ex. Suppose you implement a program that simulates solving a Rubik's cube
- When someone uses your program:
 - What abstract problem space are they interacting with?
 - What are the concrete ways they interact with the program?



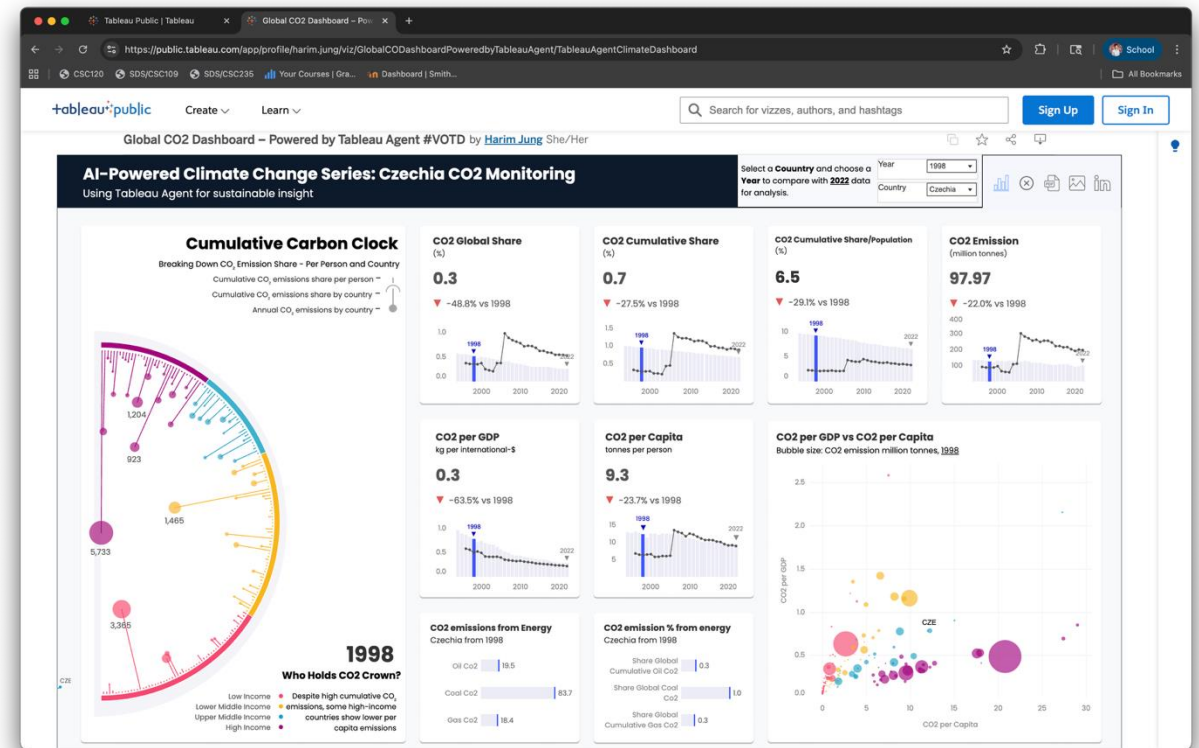
SCIENCE OF INTERACTION

- Ex. Use DataVoyager to find two variables in the Movies dataset with a negative correlation
- When you perform this task:
 - What abstract problem space are you interacting with?
 - What are the concrete ways you interact with DataVoyager?



SCIENCE OF INTERACTION

- Ex. Use Global CO2 Dashboard to find an insight about CO2 emissions in China
- When you perform this task:
 - What abstract problem space are you interacting with?
 - What are the concrete ways you interact with the dashboard?



SCIENCE OF INTERACTION

- Interaction as a Reasoning Aid
 - Interactive manipulation of an interface happens in service of the abstract problem space
 - Ex. I need to rotate parts of a virtual Rubik's Cube to solve it
 - concrete action
 - abstract problem
 - Ex. I need to select and drag variables in DataVoyager to analyze correlations
 - concrete action
 - abstract problem
 - Ex. I need to hover over data points in Global CO2 Dashboard to find China's emissions
 - concrete action
 - abstract problem

SCIENCE OF INTERACTION

- Interaction as a Reasoning Aid
 - Interactive manipulation of an interface happens in service of the abstract problem space
 - Interaction is part of the reasoning process; it is an inquiry
 - Insight emerges out of the experience of manipulating information; the more ways a user can ‘hold’ their data (ex. exploring from different angles, changing the data’s form) the more insight will accumulate
 - Acts of conceptual manipulation are called **situation**: the joining of background context and current observations, and analyses towards a goal

SCIENCE OF INTERACTION

- Interaction as a Reasoning Aid: situation
 - Lemke calls situation an 'ecology', evoking the dynamic interaction between concepts and thinkers in the process of knowledge construction
 - Situated cognition theorizes cognition as an emergent property of interactions between people and their environment through perception and action
 - In other words... cognition is not a purely mental activity; it requires interaction with artifacts
 - Interaction supports dialogical inquiry; the user and tool each poses questions and answers, as the user interacts to answer one question, they may encounter new ones

DEFINING INTERACTION FOR VISUALIZATION

Interaction for visualization is the interplay between a person and a data interface involving a data-related intent, at least one action from the person and an interface reaction that is perceived as such.

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Interaction?

Joe is staring at a visualization without performing any action that is interpretable by the visualization. He is reasoning (mentally) about the visualized data. There are further (computer initiated) animations, real time changes that show important information that Joe takes into account in his reasoning process.

E. Dimara and C. Perin, "What is Interaction for Data Visualization?," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 1, pp. 119-129, Jan. 2020, doi: 10.1109/TVCG.2019.2934283.

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Interaction?

Joe wants to focus on a cluster of data points. Using a mouse, he traces a lasso to select the data points. The visualization draws the lasso with the exact same color as the background.

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Interaction?

Joe notices a slider widget. He intends to filter out some data points. He moves the slider and the visualization is zoomed-in. He does not understand what is going on.

DEFINING INTERACTION FOR VISUALIZATION

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Interaction?

An airport provides a visualization that updates automatically to show all departure gate data. Jack looks for his gate number, but the visualization changes too quickly. A gaze detector tracks Jack's unfocused gaze attempts and slows down the rate at which the visualization changes. Jack now locates his gate number.

E. Dimara and C. Perin, "What is Interaction for Data Visualization?," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 1, pp. 119-129, Jan. 2020, doi: 10.1109/TVCG.2019.2934283.

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Interaction?

Averell cannot remember where he parked his car. In the parking a screen shows car CO2 emission data to motivate car drivers to use their car less. Averell is not careful and he bumps his head on the screen. Because the screen is touch-enabled, this contact rearranges the data points. Averell sees some graphics moving around but does not pay attention. He ignores the screen and continues his search.

E. Dimara and C. Perin, "What is Interaction for Data Visualization?," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 1, pp. 119-129, Jan. 2020, doi: 10.1109/TVCG.2019.2934283.

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Interaction?

Like Averell, William bumps his head on the display. But when he sees the graphics move around he gets curious and starts touching the display to explore what these graphic things are about.

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LEVERAGING INTERACTION

- Recall: Interaction is a reasoning aid to the user; it enables situated cognition
- Question: Can interaction also benefit the visual interface? How?

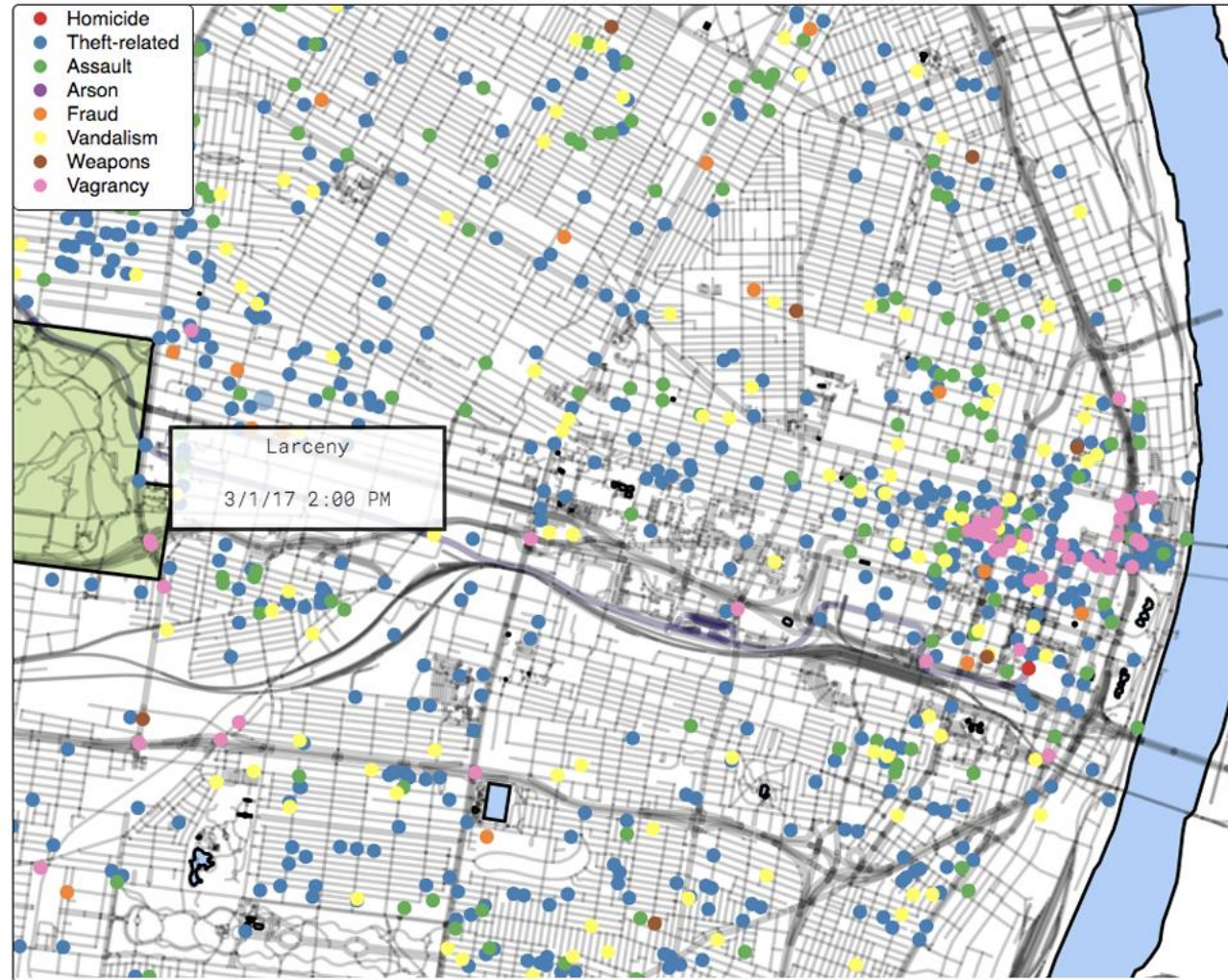
LEVERAGING INTERACTION

- Predicting User Intent
 - A visual interface can track user interactions and use that data to build predictive models of user goals and needs.
 - Given an accurate prediction of user goals and needs, an interface can proactively update to support them.

LEVERAGING INTERACTION

- Predicting User Intent:
Ex. Inferring Interest
based on Clicks

Ottley, A., Garnett, R. and Wan, R. (2019), Follow The Clicks: Learning and Anticipating Mouse Interactions During Exploratory Data Analysis. Computer Graphics Forum, 38: 41-52. <https://doi.org/10.1111/cgf.13670>



LEVERAGING INTERACTION

- Illuminating Potential Bias
 - Bias in data analysis can look like over-analysis of some subset(s) of data and under-analysis of others.
 - Interaction can illuminate patterns of exploration and highlight over-/under-analysis patterns.

LEVERAGING INTERACTION

■ Illuminating Potential Bias: Ex. Lumos

Arpit Narechania, Adam Coscia, Emily Wall, and Alex Endert. 2022. Lumos: Increasing Awareness of Analytic Behavior during Visual Data Analysis. IEEE Transactions on Visualization and Computer Graphics 28, 1 (Jan. 2022), 1009–1018. <https://doi.org/10.1109/TVCG.2021.3114827>



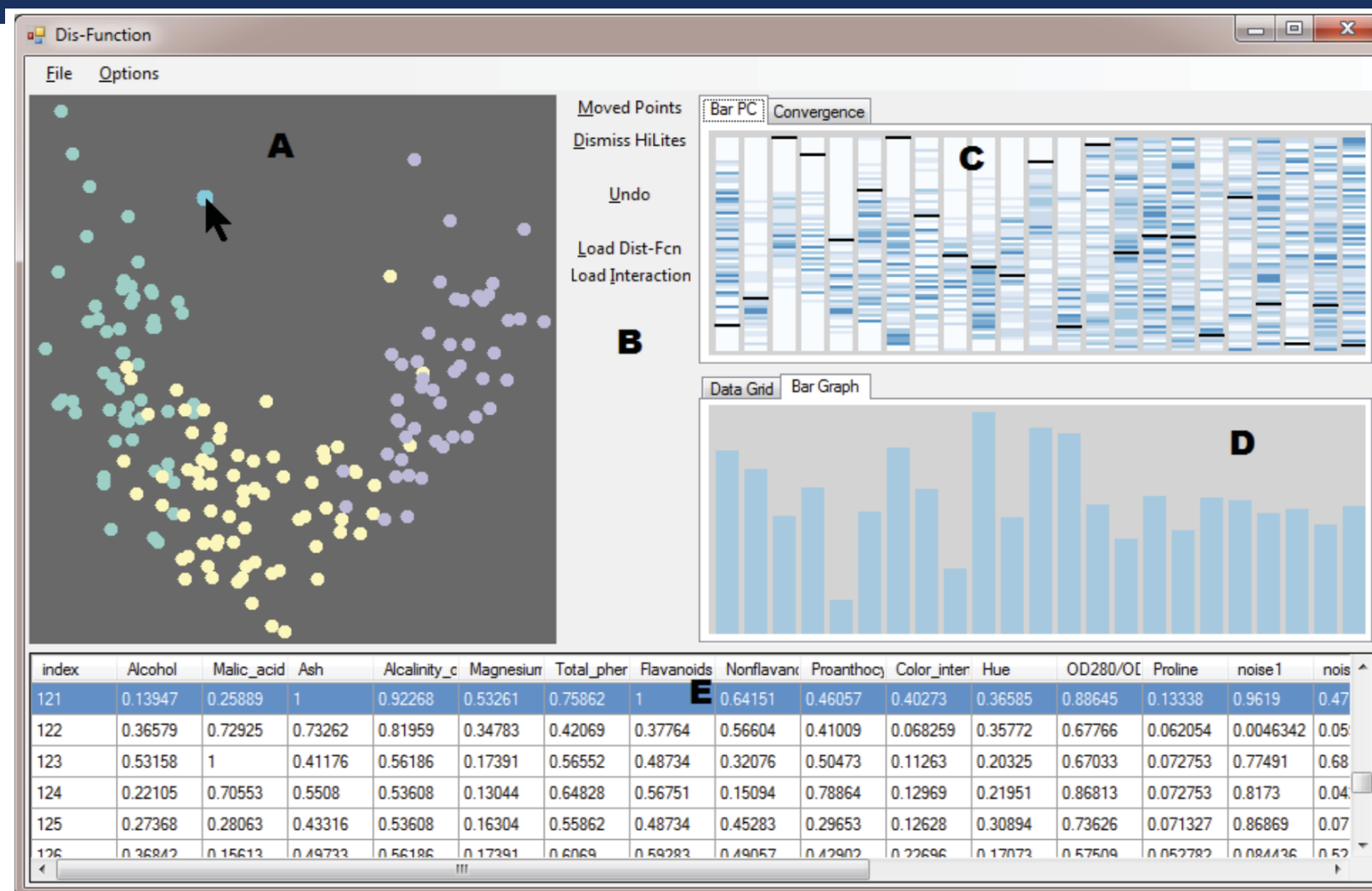
LEVERAGING INTERACTION

- Update Underlying Models
 - Advanced modeling is used to analyze data, subject matter experts can evaluate analysis results for anomalies and mistakes, but may not have the technical skills to tune a model
 - Interaction can be used to allow subject matter experts to tune model parameters without requiring technical skills

LEVERAGING INTERACTION

- Update Underlying Models: Ex. Dis-Function

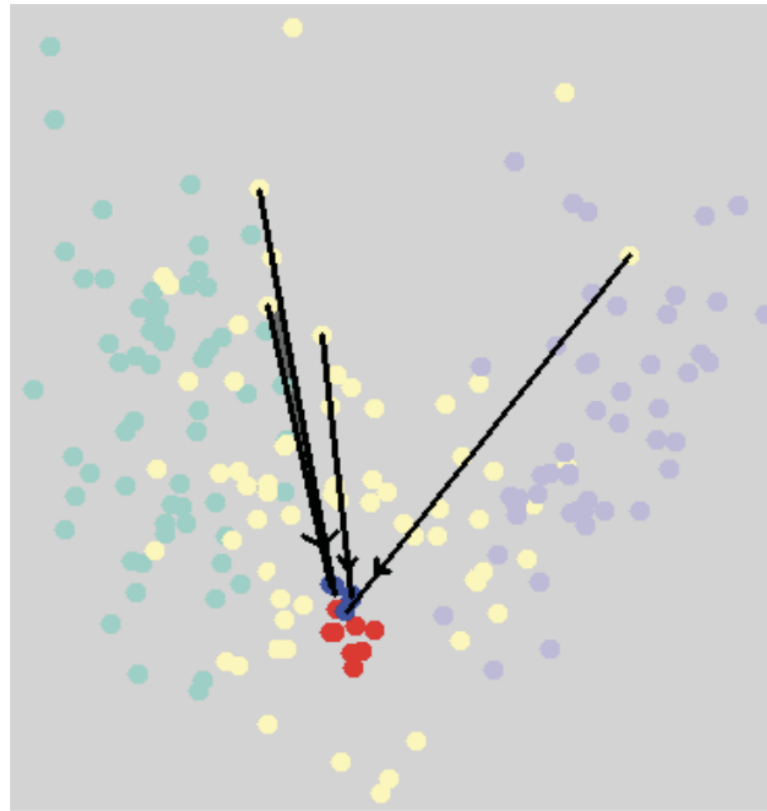
E.T. Brown, J. Liu, C. E. Brodley and R. Chang, "Dis-function: Learning distance functions interactively," *2012 IEEE Conference on Visual Analytics Science and Technology (VAST)*, Seattle, WA, USA, 2012, pp. 83-92, doi: 10.1109/VAST.2012.6400486.



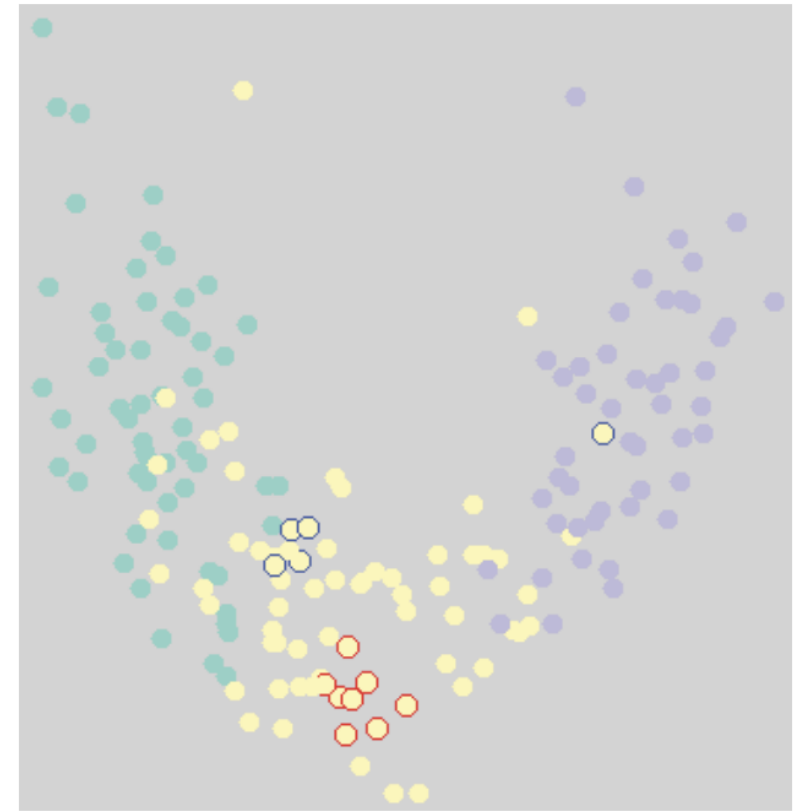
LEVERAGING INTERACTION

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(a)



(b)

LEVERAGING INTERACTION

- Question: If you were to build a system that tracks user interactions, how might you do it? What data would you collect?